Soil sample DNA extraction (SuperSoil)

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Abstract

A method for fungal and bacterial DNA extraction.

Safety warnings

1. Solution SEA and Ethanol are flammable
2. DO NOT add bleach or acidic solutions directly to the sample preparation waste.

Materials

1. Bead (ball mill)

Zirconia beads	0.5	6	0.2	CAAR-ZB-0.5-1000BX
Zirconia beads	1.0	6	0.2	CAAR-ZB-1-1000BX

2. Solution SD1 (Chaotropic)

	w/w
sodium thiocyanate	2.5%
Na_2HPO_4	5%
pH	~8.4

3. Solution SD2

	w/w
$AlCl_3 \cdot 6H_2O$	1.5%
Ammonium acetate	25%
pH (Adjust with acetic acid)	6.3

4. Solution SD3 (Chaotropic)

	w/w
Guanidinium thiocyanate	65%
pH	~7.0

Solution SEA (Wash buffer 1)

	w/w
Ethanol	50%
Isopropanol	10%
Guanidine hydrochloride	30%

80% Ethanol (Wash buffer 2)

DEPC treated water (Elution buffer)

Procedure

Sample preparation & Cell lysis

Step 1

Add **0.2 mL** ⌀=1.0 mm zirconia beads, **0.2 mL** ⌀=0.5 mm zirconia beads, **800 µL** SD1 solution, **0.5 g** sample into a 1.5 ml microcentrifuge tube. Vortex briefly to mix.

> **Note**
>
> If the SD1 solution is precipitated, leave it at room temperature for 30 minutes, it will become transparent again.

> **Note**
>
> If the soil is too loose and fills up the whole microcentrifuge tube, you can separate them into 2 microcentrifuge tubes for the downstream workflow. And centrifuge 2 microcentrifuge tubes at the final step to merge them.

> **Note**
>
> If your sample is hard to lyse, you can try the following methods:
>
> 1. After adding solution SD1, incubate at 65 ℃ for 10 minutes
> 2. Add 3 additional stainless steel beads and use 2.0 ml screw vials for your samples

Step 2 *(4 min)*

Homogenize samples using Vortex-Genie® 2 with SI-H524 Horizontal microtube holder. Vortex at maximum speed for **4 min**

> **Note**
>
> Be careful when removing the microcentrifuge tube from the holder. Press the lid and slowly loosen the tube with another hand.

Step 3 *(2 min)*

Centrifuge the 1.5 ml microcentrifuge tube at**14000 rpm, 25 °C, 2 min**

Inhibitor precipitation

Step 4

Add **200 µL** SD2 solution to a clean 1.5 ml microcentrifuge tube.

Step 5 *(5 s)*

Transfer **500 µL** supernatant to the tube in the previous step. Vortex **5 s** for mixing.

> **Note**
>
> White precipitate forms.
>
> If too much AlCl3 (CD2) is added, it will cause all the DNA to be precipitated along with the inhibitory substances.

> **Note**
>
> If there is any suspended substance, you can use P200 and turn it to 150 µl and suck the supernatant up 3 times.

Step 6 *(2 min)*

Centrifuge at**14000 rpm, 25 °C, 2 min**

DNA binding

Step 7 *(5 s)*

Add **600 µL** SD3 solution to a clean 1.5 ml microcentrifuge tube. Avoiding the pellet, transfer up to **700 µL** of supernatant from the previous step to that tube. Vortex **5 s** for mixing.

> **Note**
>
> If Solution CD3 has precipitated, heat at 60°C until the precipitate dissolves.

Step 8 *(2 min)*

Take **650 µL** to spin column (with collection tube) and centrifuge at**14000 rpm, 25 °C, 2 min**

Step 9

Discard the flow-through and repeat step 8 to ensure all the lysates have passed through the spin column.

Step 10

Carefully place the spin column into a clean 2 ml collection tube. Avoid splashing any flow-through onto the spin column.

Wash

Step 11 *(2 min)*

Add **500 µL** of Solution SEA to the spin column. Centrifuge at **14000 rpm, 25 °C, 2 min**

> **Note**
>
> Solution SEA may leaks or suck back causing contamination.

Step 12

Discard the flow-through and return the spin column to the same 2 ml collection tube.

Step 13 *(2 min)*

Add **500 µL** 80% ethanol to the spin column. Centrifuge at **14000 rpm, 25 °C, 2 min**

> **Note**
>
> Ethanol may leaks or suck back causing contamination.

Step 14

Discard the flow-through and place the spin column into a new 2 ml collection tube.

Step 15 *(3 min)*

Centrifuge at **14000 rpm, 25 °C, 3 min** Carefully place the spin column into the new 1.5 ml microcentrifuge tube

> **Note**
>
> Ethanol can interfere with downstream DNA applications, such as PCR, restriction digests, and gel electrophoresis. Make sure to centrifuge for 3 minutes.

Elute

Step 16

Add **50 µL** RNase-free water (DEPC-treated water) to the center of the white filter membrane.

> **Note**
>
> Alternative for RNase-free water: TE(Tris-EDTA) buffer

Step 17 *(2 min)*

Centrifuge at **14000 rpm, 25 °C, 2 min** Discard the spin column. The DNA will be in the microcentrifuge tube.

Quality and quantity assessment

Step 18

Use BioTek Take3 & Take3 Trio Microvolume Plates to measure the quality and quantity of your specimens. And then upload them to Google Drive (1 MolecularWork (Direct edit) 分生實驗號碼 (直接編輯) w_ Primer list)